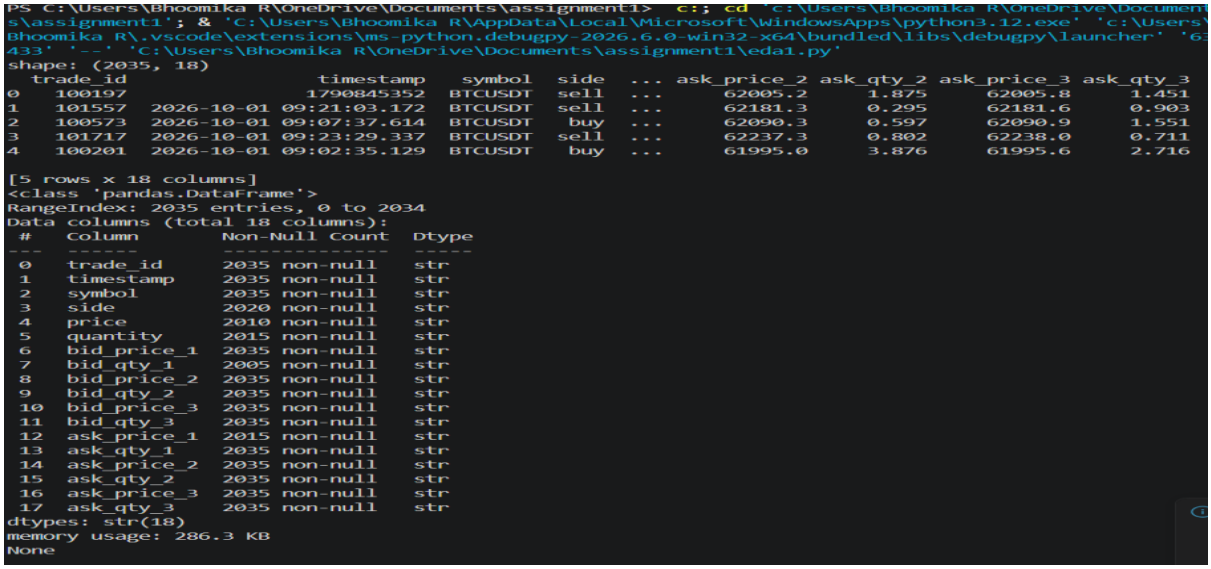
Results:-

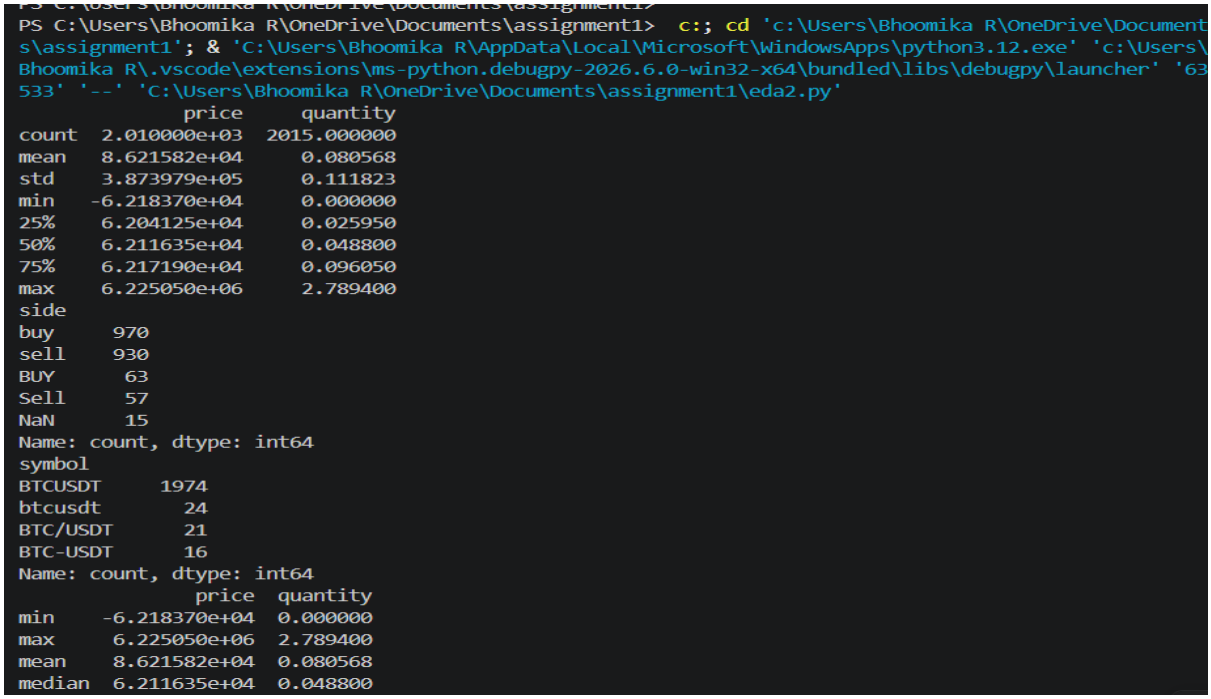
Step 1:-



```
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>  c:; cd 'c:\Users\Bhoomika R\OneDrive\Document
s\assignment1'; & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63
433' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda1.py'
shape: (2035, 18)
  trade_id                timestamp    symbol   side  ... ask_price_2 ask_qty_2 ask_price_3 ask_qty_3
0   100197                1790845352  BTCUSDT   sell  ...      62005.2      1.875      62005.8      1.451
1   101557  2026-10-01 09:21:03.172  BTCUSDT   sell  ...      62181.3      0.295      62181.6      0.903
2   100573  2026-10-01 09:07:37.614  BTCUSDT    buy  ...      62090.3      0.597      62090.9      1.551
3   101717  2026-10-01 09:23:29.337  BTCUSDT   sell  ...      62237.3      0.802      62238.0      0.711
4   100201  2026-10-01 09:02:35.129  BTCUSDT    buy  ...      61995.0      3.876      61995.6      2.716

[5 rows x 18 columns]
<class 'pandas.DataFrame'>
RangeIndex: 2035 entries, 0 to 2034
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   trade_id       2035 non-null   str
 1   timestamp      2035 non-null   str
 2   symbol         2035 non-null   str
 3   side           2020 non-null   str
 4   price          2010 non-null   str
 5   quantity       2015 non-null   str
 6   bid_price_1    2035 non-null   str
 7   bid_qty_1      2005 non-null   str
 8   bid_price_2    2035 non-null   str
 9   bid_qty_2      2035 non-null   str
 10  bid_price_3    2035 non-null   str
 11  bid_qty_3      2035 non-null   str
 12  ask_price_1    2015 non-null   str
 13  ask_qty_1      2035 non-null   str
 14  ask_price_2    2035 non-null   str
 15  ask_qty_2      2035 non-null   str
 16  ask_price_3    2035 non-null   str
 17  ask_qty_3      2035 non-null   str
dtypes: str(18)
memory usage: 286.3 KB
None
```

Step 2:-



```
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>  c:; cd 'c:\Users\Bhoomika R\OneDrive\Document
s\assignment1'; & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63
533' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda2.py'
              price     quantity
count  2.010000e+03  2015.000000
mean   8.621582e+04     0.080568
std    3.873979e+05     0.111823
min   -6.218370e+04     0.000000
25%    6.204125e+04     0.025950
50%    6.211635e+04     0.048800
75%    6.217190e+04     0.096050
max    6.225050e+06     2.789400
side
buy     970
sell    930
BUY      63
Sell     57
NaN      15
Name: count, dtype: int64
symbol
BTCUSDT     1974
btcusdt       24
BTC/USDT      21
BTC-USDT      16
Name: count, dtype: int64
              price  quantity
min    -6.218370e+04  0.000000
max     6.225050e+06  2.789400
mean    8.621582e+04  0.080568
median  6.211635e+04  0.048800
```

Step 3:-

```
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>  c:; cd 'c:\Users\Bhoomika R\OneDrive\Document
s\assignment1'; & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63
619' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda3.py'
trade_id        0
timestamp       0
symbol          0
side           15
price          25
quantity       20
bid_price_1     0
bid_qty_1      30
bid_price_2     0
bid_qty_2       0
bid_price_3     0
bid_qty_3       0
ask_price_1    20
ask_qty_1       0
ask_price_2     0
ask_qty_2       0
ask_price_3     0
ask_qty_3       0
dtype: int64
duplicate rows: 35
duplicate trade_id: 35
timestamp
9999-99-99 99:99:99.999    1917
99/99/9999 99:99:99          78
9999999999                   40
Name: count, dtype: int64
price <= 0: 6
quantity <= 0: 6
crossed books: 10
```

Step 4:-

```
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>  c:; cd 'c:\Users\Bhoomika R\OneDrive\Document
s\assignment1'; & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63
718' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda4.py'
clean rows: 1883
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>
```

Step 5:-

```
TypeError: unsupported operand type(s) for -: 'str' and 'str'
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1> & "C:\Users\Bhoomika R\AppData\Local\Microso
\WindowsApps\python3.12.exe" "c:/Users/Bhoomika R/OneDrive/Documents/assignment1/eda5.py"
Q1: 62042.65 Q3: 62171.6 IQR: 128.9499999999971
fences: 61849.225000000006 62365.024999999994
outliers found: 0
Empty DataFrame
Columns: [trade_id, timestamp, price]
Index: []
rows after outlier removal: 1883
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>
```

Step 6:-

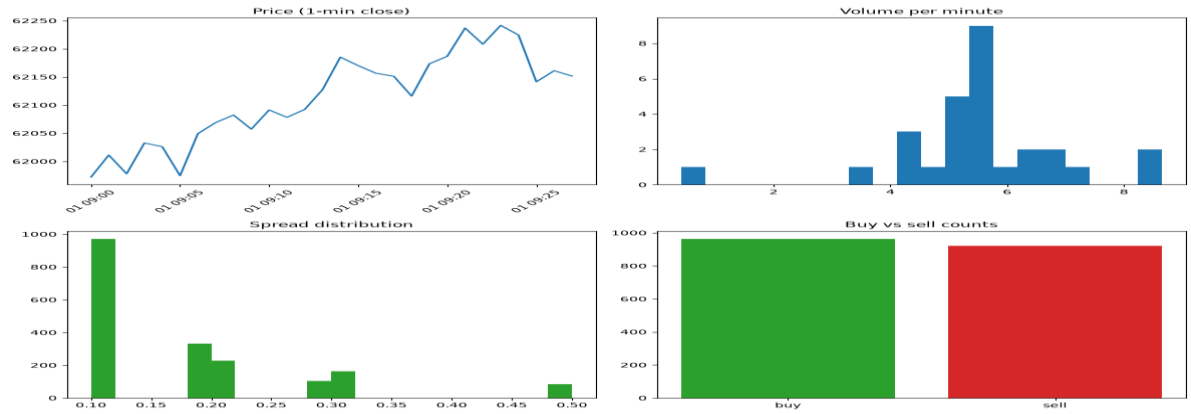
```
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1> & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '53068' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda6.py'
              spread     mid_price  order_imbalance  ...  volatility_20         ma_10       rsi_14
count  1883.000000   1883.000000      1883.000000  ...    1863.000000   1874.000000  1869.000000
mean      0.176208  62109.198672        -0.032684  ...       0.000067  62109.375374    51.332511
std       0.100409     80.265917         0.526706  ...       0.000011     80.010941    16.834481
min       0.100000  61954.050000        -0.979640  ...       0.000033  61961.950000     7.538803
25%       0.100000  62042.600000        -0.478600  ...       0.000058  62042.345000    39.009901
50%       0.100000  62116.650000        -0.036335  ...       0.000067  62118.275000    51.941748
75%       0.200000  62171.550000         0.417966  ...       0.000075  62171.615000    63.981043
max       0.500000  62253.950000         0.986384  ...       0.000095  62250.310000    94.642857

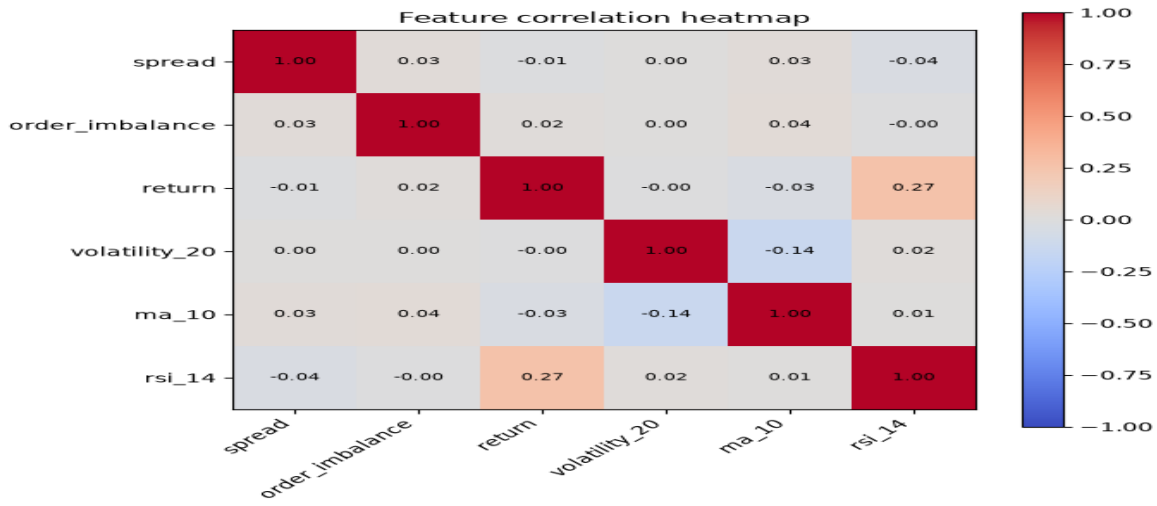
[8 rows x 7 columns]
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>
```

Step 7:-

```
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '51361' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda7.py'
                        open     high      low    close  volume  trade_count
timestamp
2026-10-01 09:00:00  61994.6  62006.7  61962.9  61972.0  6.2213           71
2026-10-01 09:01:00  61969.0  62030.4  61967.2  62011.2  7.1942           73
2026-10-01 09:02:00  62019.6  62020.4  61976.7  61977.9  6.7726           70
2026-10-01 09:03:00  61978.9  62039.0  61954.1  62032.7  5.3915           79
2026-10-01 09:04:00  62028.7  62062.9  62020.1  62026.1  4.2172           56
number of candles: 28
total volume: 153.9892
total quantity in trades: 153.9892
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>
```

Step 8:-





Step 9:-

```
s\assignment1'; & 'C:\Users\Bhoomika R\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users
Bhoomika R\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '5
863' '--' 'C:\Users\Bhoomika R\OneDrive\Documents\assignment1\eda9.py'
rows: 16 | train: 11 | test: 5
train period: 2026-10-01 09:07:00 -> 2026-10-01 09:17:00
test period:  2026-10-01 09:18:00 -> 2026-10-01 09:22:00
share up in train: 0.818
share up in test:  0.4
PS C:\Users\Bhoomika R\OneDrive\Documents\assignment1>
```